# Master’s thesis workflow

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Recruiter portfolio case study
R + Stata | Dynamic panel modelling

## How does Tourism affect Housing Affordability across Austrian Districts?

A district-level panel analysis of Austria from 2015-2024.

STATcube and OeNB as main datasources.

# One policy question, two competing channels

RESEARCH QUESTION

## Does more tourism make local housing less affordable?

$$HAI_{it} = \frac{Net\ income\ pc_{it}\ (Y_{it})}{Median\ house\ price\ per\ m^2_{it}\ (P_{it})},$$

*(HAI = Housing Affordability Index)*

### Demand pressure

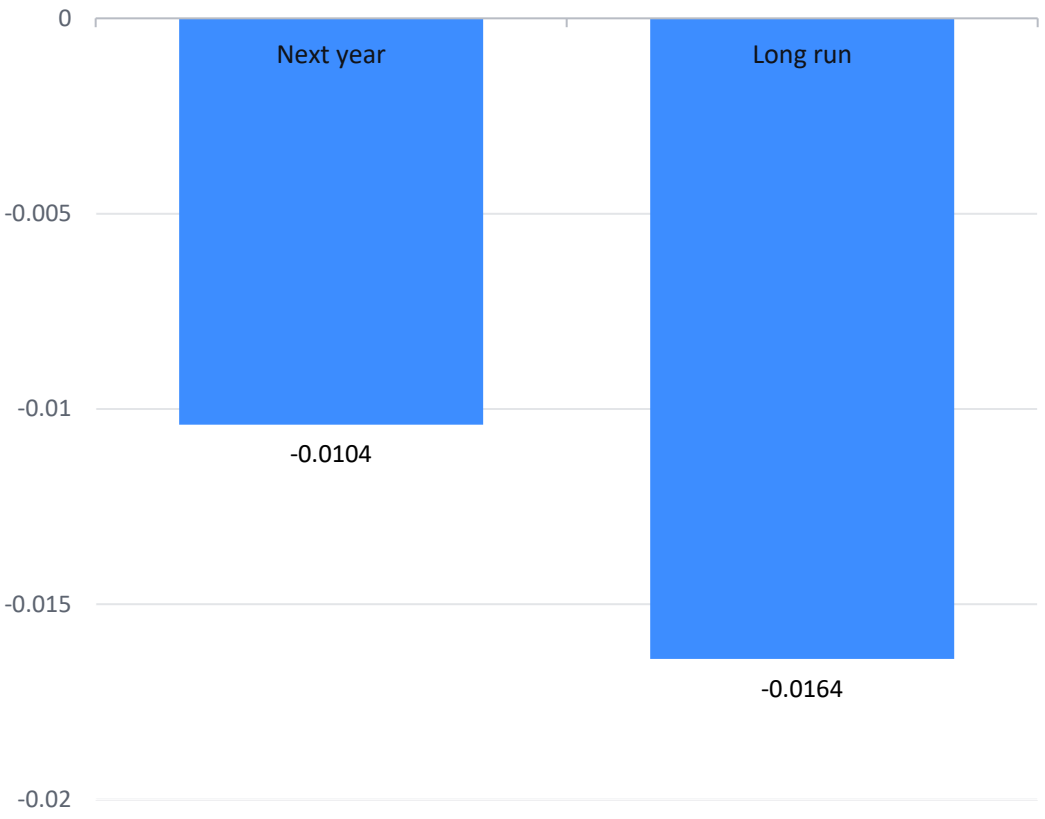
Tourist demand, property investment, improved amenities and second home demand can increase residential prices.

### Local income gains

Employment and business activity can raise purchasing power and offset price effects.

**The empirical task was to estimate the net effect - then test where it is strongest and which component drives it.**

# Main result and policy interpretation




Effect of a 10% increase in tourism intensity

## Main finding

A 10% increase in tourism intensity is associated with approximately 1.0% lower housing affordability in the following year.

## Where the relationship is clearest

The effect is stronger in highly tourism-dependent districts and clearer for prices of houses than for flats.

## Policy implication

Retain the regional economic benefits of tourism, but address housing pressures specifically in highly tourism-dependent destinations.

# I built a reliable district-year panel

The engineering challenge was harmonisation: changing district definitions, mixed file shapes, suppressed values and measurement breaks.

01

## Ingest

Tourism, prices, income, population, migration, dwellings and labour market data

02

## Standardise

Parse district codes, pivot wide/long formats, normalise units and consolidate Vienna

03

## Engineer

Create affordability, tourism intensity, lags, logs, growth rates and supply controls

04

## Validate

Check keys and duplicates, align coverage, inspect missingness and export analysis panels

**94** districts in raw panel

**940** potential district-years

**822** preferred-model observations

# Code case study: cleaning Statcube exports

R | RESHAPE AND PARSE PUBLIC-DATA EXPORTS – SIMPLIFIED EXTRACT FROM CODE SCRIPT

```
tourism_clean <- hotel %>%
  mutate(
    district_code = str_extract(
      District, "(?<=<)\\d+(?=>)") |> as.integer(),
    district = str_remove(District, "\\s*<\\d+>$")
  ) %>%
  pivot_longer(
    matches("^(Winter-half|Summer-half)"),
    names_to = "period",
    values_to = "overnight_stays",
    values_transform =
      list(overnight_stays = as.character)
  ) %>%
  mutate(
    overnight_stays = parse_number(
      str_trim(overnight_stays),
      na = c("", "-", "–", "..", "...", ":", "NA")),
    season = if_else(str_detect(period, "Winter"),
                     "winter", "summer"),
    year = str_extract(period, "\\d{4}") |> as.integer()
  )
```

## Problem

Statcube exports combined district names and codes, stored seasons in separate columns and used publication symbols in numeric fields.

## Solution

Regular expressions separated district identifiers. pivot_longer created a consistent row structure, while parse_number mapped suppression markers to missing values.

## Why it matters

The pipeline kept missing observations distinct from zero and produced comparable district-year-season records.

# Code case study: repairing a reporting break

R | ALLOCATION AND RECONCILIATION

```
income3_allocated <- income3_grouped %>%
  left_join(income_2022_weights,
            by = c("district_group", "value_type"),
            relationship = "many-to-many") %>%
  group_by(year, value_type, district_group) %>%
  arrange(district_code, .by_group = TRUE) %>%
  mutate(
         allocated_raw = grouped_total * share_2022,
         value = round(allocated_raw),
         value = if_else(row_number() == n(),
         value + grouped_total - sum(value), value),
         income_estimated = TRUE) %>%
  ungroup()
```

## Problem

Income data merged five districts in 2023-24, breaking district-level comparability.

## Solution

Use the last observed district shares, allocate grouped totals, then force the rounded parts to reconcile exactly to the published aggregate.

## Why it matters

The workflow preserves the published totals and flags the allocated district values as estimates.

# Code case study: variable construction

R | FEATURE ENGINEERING FOR THE PANEL

```
# Create main explanatory variables
tourism_panel <- tourism_yearly %>%
  left_join(population, by = c("district_code","year")) %>%
  left_join(country_total, by = "year") %>%
  mutate(tourism_intensity_ratio = (overnight_stays/ population) /
        overnight_share_country) %>%
  mutate(tourism_intensity = overnight_stays / population)
log_tourism <- tourism_panel %>%
  mutate(log_tourism_intensity = log(tourism_intensity),
        log_tourism_intensity_ratio = log(tourism_intensity_ratio),
        log_overnight_stays = log(overnight_stays)
)

# Create main dependent variable
housing_affordability <- income_entity %>%
  left_join(price_house, by = c("district_code","year")) %>%
    arrange(district_code,year) %>%
    group_by(district_code) %>%
  mutate(housing_affordability_index = income_pc/price_house ,
        log_housing_affordability_index = log(income_pc/price_house)
)
```

## Exposure variable

Overnight stays per resident captures tourism pressure while making districts of different sizes comparable.

## Outcome variable

Income per resident divided by median house price per m² converts prices into an interpretable affordability measure.

## Model-ready transformations

Logs, lags, growth rates and supply-control ratios improve comparability and align the variables with the econometric design.

# Main econometric specification

$$log(HAI_{it}) = \rho\, log(HAI_{i,t-1}) + \beta\, log(TI_{i,t-1}) + X'_{it}\gamma + \mu_i + \lambda_t + \varepsilon_{it}$$

### Dynamic outcome

Affordability shocks carry forward, a lagged dependent variable is required to capture persistence.

### Lagged explanatory variable

A one-year lag of tourism reduces simultaneity, and considers that the housing market reacts slowly.

### District and year effects

District fixed effects (μ) absorb stable local characteristics. Year fixed effects (λ) capture shocks shared across Austrian districts.

### Control variables (X')

Controls reflect economic theory and capture relevant time-varying conditions within districts.

### Instrument discipline

Collapsed, bounded lag instruments limit proliferation and match each variable's assumed timing.

### Model checks

AR(2), Hansen and Difference-in-Hansen tests assess serial correlation and instrument validity.

**Main econometric challenge:**
**Endogeneity from reverse causality, persistent housing affordability and unobserved district characteristics**

**Solution: Two-step Blundell-Bond System GMM estimator using internal lag instruments**

# Code case study: disciplined System GMM estimation

R | SYSTEM GMM CROSS-CHECK

```
pgmm_m6 <- pgmm(
  log_housing_affordability_index ~
  lag(log_housing_affordability_index, 1) +
  lag(log_tourism_intensity, 1) + log_density +
  unemployment_rate  + log_dwellings_pc  +
  new_dwellings_share + migration_share |
    lag(log_housing_affordability_index, 2:9) +
    lag(log_tourism_intensity, 2:9) +
    lag(log_density, 1:2) + lag(log_dwellings_pc, 1:2) +
    lag(unemployment_rate, 2:9) + lag(new_dwellings_share,
    2:9) + lag(migration_share, 2:9),
  data = pdata_agg,
  effect = "twoways",
  model = "twosteps",
  transformation = "ld",
  collapse = TRUE
)
summary(pgmm_m6,robust = TRUE,time.dummies = TRUE)
```

## 1 Separate timing assumptions

Tourism, migration, unemployment and construction are endogenous; instrumented on deeper lags. Density and housing stock are predetermined; instrumented on lags from previous year.

## 2 Control instrument count

Collapsed instruments and bounded lag windows reduce instrument proliferation in a short panel.

## 3 Triangulate implementations

Static and diagnostic work ran in R; the final xtabond2 estimates in Stata were replicated with R's pgmm specification as a robustness check.

**Result: the Stata and R implementations produced closely aligned preferred estimates and instrument counts.**

# Robustness testing and model validation

Each check targeted a specific threat to inference rather than adding specifications without a clear purpose.

### 01 Diagnose the panel

Test heteroskedasticity, serial correlation and cross-sectional dependence before selecting the estimator.

### 02 Stress-test assumptions

Compare static and dynamic models, alternative lag timing and bounded instrument windows.

### 03 Change measures and samples

Use tourism levels and growth, house and flat outcomes, high-tourism subsamples and COVID exclusions.

### 04 Validate the preferred model

Review AR(2), Hansen and Difference-in-Hansen tests, limit instruments and cross-check R against Stata.

**Preferred-model diagnostics: 822 observations | 93 districts | 60 instruments | AR(2) p = 0.851 | Hansen p = 0.112**

# Testing snow as an instrument for winter tourism

Weather-station observations were converted into a district-year instrument, then used in an exploratory two-stage SLS estimator.

RECONSTRUCTED SF MATCH + SUBMITTED IV MODEL

```
library(sf)
stations_sf <- st_as_sf(
  stations, coords = c("Länge [°E]", "Breite [°N]"),
  crs = 4326) |> st_transform(3035)
district_points <- district_zones |>
  st_transform(3035) |> st_point_on_surface()
nearest <- st_nearest_feature(district_points, stations_sf)
district_station <- district_points |>
  mutate(station = stations_sf$id[nearest]) |>
  st_drop_geometry()
snow_iv <- district_station |>
  left_join(station_year, by = "station") |>
  select(district_code, year, snow_days_30cm)
fullpanel_winter_iv <- fullpanel_winter_iv |>
  left_join(snow_iv, by = c("district_code", "year"))
iv_winter <- feols(
  log_housing_affordability_index ~ "controls" |
  district_code + year |
  log_winter_tourism ~ snow_days_30cm,
  data = fullpanel_winter_iv,
  cluster = ~district_code)
summary(iv_winter)
```

## 1 Locate stations

456 metadata records provide geotags; 75 station IDs appear in the matched panel.

## 2 Match spatially

The output indicates nearest-station matching: one station can serve several districts, picked the one closest to 1000 masl for comparability.

## 3 Build the instrument

Quality-screened snow measures produce 940 district-years: 94 districts over 2015-2024.

## 4 Test relevance

The first stage links snow to winter tourism; weak-instrument diagnostics accompany 2SLS.

**Snow provides external variation in winter tourism, but causality requires that it affects affordability only through tourism.**

# Skills demonstrated across the research workflow

**Data engineering**
Cleaned public extracts, reconciled reporting breaks, joined district panels and engineered analytical variables.

**Econometrics**
Diagnosed panel structure, estimated System GMM, OLS and IV models, and challenged assumptions with robustness checks.

**Communication**
Translated code, diagnostics and evidence into a concise business and policy narrative.

## Analytical judgement

I report the association and exploratory IV design with their limits: instrument assumptions, a ten-year panel, COVID disruption, district aggregation and missing rental/property-level data.

Portfolio case study | Source package: thesis, R pipelines, analysis CSV and Stata log